OpenWebSearch.eu — A European Open Web Index and its potential Applications

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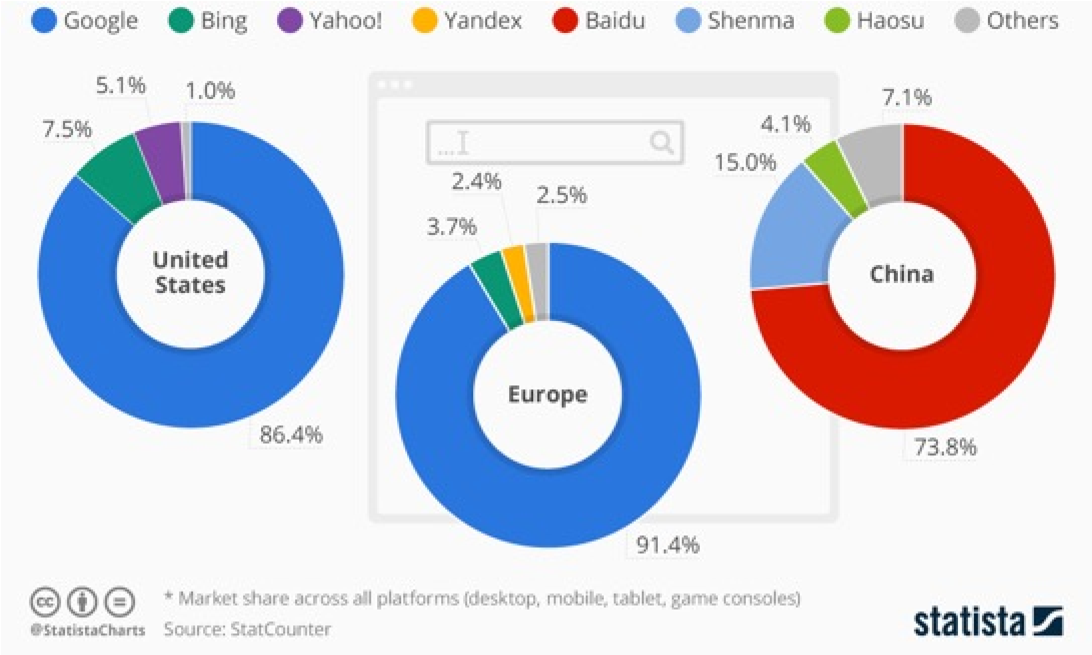
Web Search Today

A critical resource managed as an oligopoly

Two properties that don't fit

- A **critical infrastructure** for society, comparable to satellite navigation
- A **market oligopoly**: dominated by a small number of gatekeepers (Google, Microsoft, Baidu, ...)



But does it matter? Yes — on four dimensions

User Choice & Democracy

- Single gatekeeper for society's information
- Search rankings can shift voting preferences by **20–72%** (SEME)
- Monopolistic ranking = single point of failure for information quality

Societal & Security Risks

- European OSINT depends on US-controlled infrastructure
- Filter bubbles, algorithmic bias
- Misinformation & Disinformation

Untapped Richest Information Source

- 60 % of web pages carry structured data
- No open access to web-scale document collections
- Science, research and technology heavily depend on the Web as platform

AI & Innovation

- Web data is the fuel for LLMs, RAG, and agentic AI
- Closed indices block reproducible research & open innovation

> Being able to navigate the Web is as fundamental as GPS or the power grid

OpenWebSearch.eu — Our Mission

> Building an Open Index of the Web and an open infrastructure to reduce upfront costs for researchers, innovators and companies in tapping the Web as a resource for search, web-analytics and AI.

Four Objectives

1. **Open Technology Stack** Open-source pipelines for crawling, preprocessing, indexing
2. **Resource Provision** Building a network of infrastructure providers (EuroHPC / AI Factories)
3. **Added Value Services** Dashboard, tools, data access APIs
4. **Bootstrapping the Ecosystem** Third-party calls, community building, applications and use-cases

14 Core Partners

Infrastructure

Research

NGOs

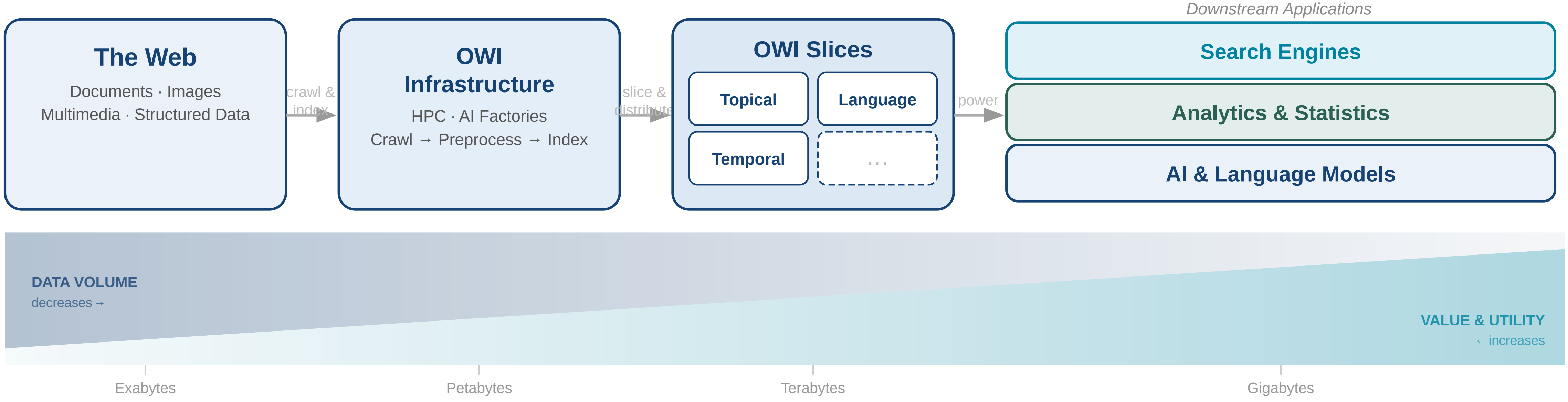
Businesses

Plus 16 financially supported third-party projects (FSTPss)

The Open Web Index (OWI)

A **Web Index** = a data structure for fast query-based access and ranking of web documents — the core of every web search engine

Our Proposition: A collaboratively created, open and transparent Web Index running on existing HPC computing centres / AI Factories in Europe



Granitzer, Michael, et al. "Impact and development of an Open Web Index for open web search." Journal of the Association for Information Science and Technology (2023).

What is the Open Web Index?

The Open Web Index

The Open Web Index — a federated, daily-published web index built across European HPC sites and consumed as composable shards.

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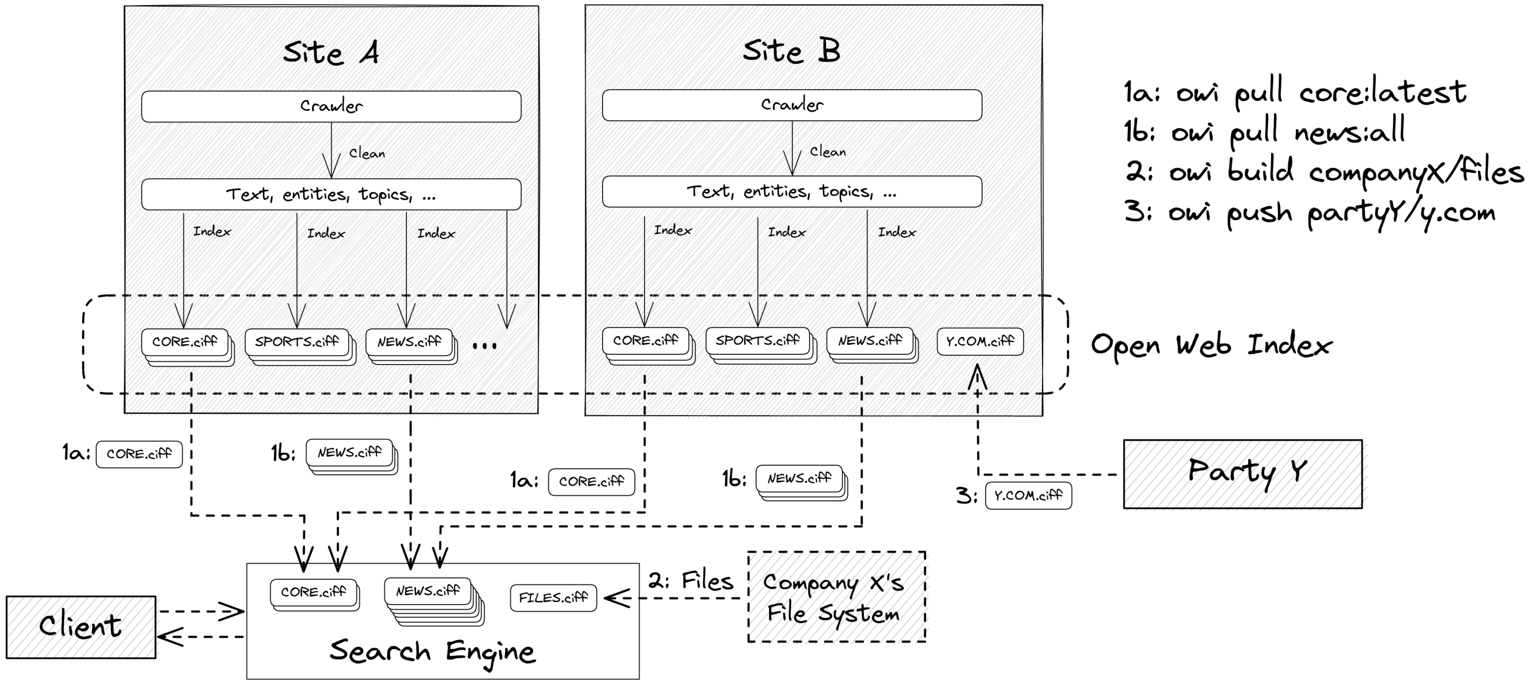

Federated production

- each HPC site runs the same pipeline independently
- daily shards by date, language, topic

Three access verbs

- `owi pull` — download pre-built shards
- `owi build` — index your own data
- `owi push` — contribute shards back

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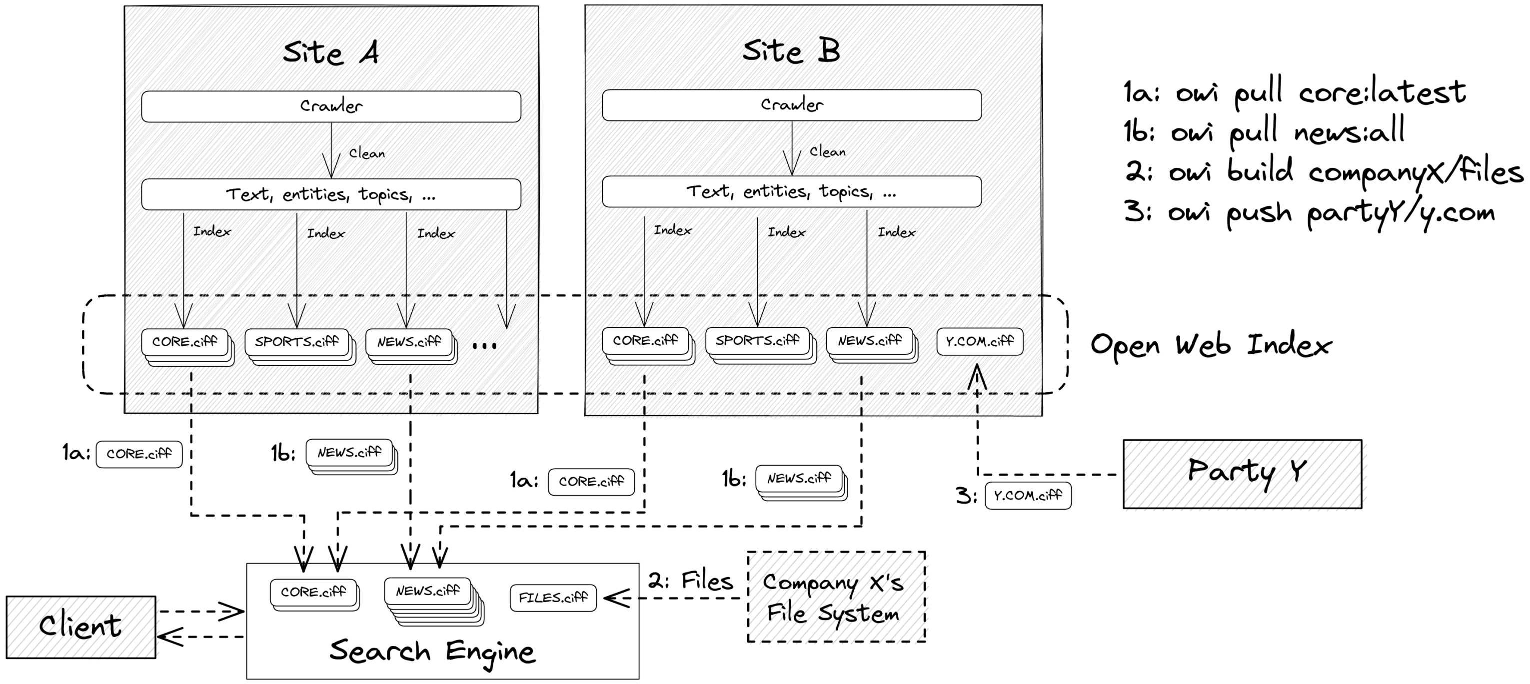

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Three output types per daily run — the building blocks of a shard:

OWI at a Glance — Statistics (Feb 2026)

Crawling

Pages crawled	~39 B total, ~12 B unique
Unique hosts	~500 M (extrapolated)
Crawling effort	1,927 machine-days
Avg. throughput	~20 M URLs / day

Index

Documents indexed	~10.7 B
Unique domains	~75 M
Public datasets	511 main / 1,511 total
Index size	45 TB (main) / 50 TB total
Embeddings	119 M docs · 884 GB

Top languages: English (38.9%), German (7.2%), Spanish (6.4%), French (5.7%), Russian (5.2%)

Multimedia Objects: Roughly 2-3 quality images and 0.5 Videos per Web-Document

Comparison: To meet commercial index size, OWI would need to scale by a factor of 20× — factor 10–50× for multimedia

Step 1: Crawling the Web

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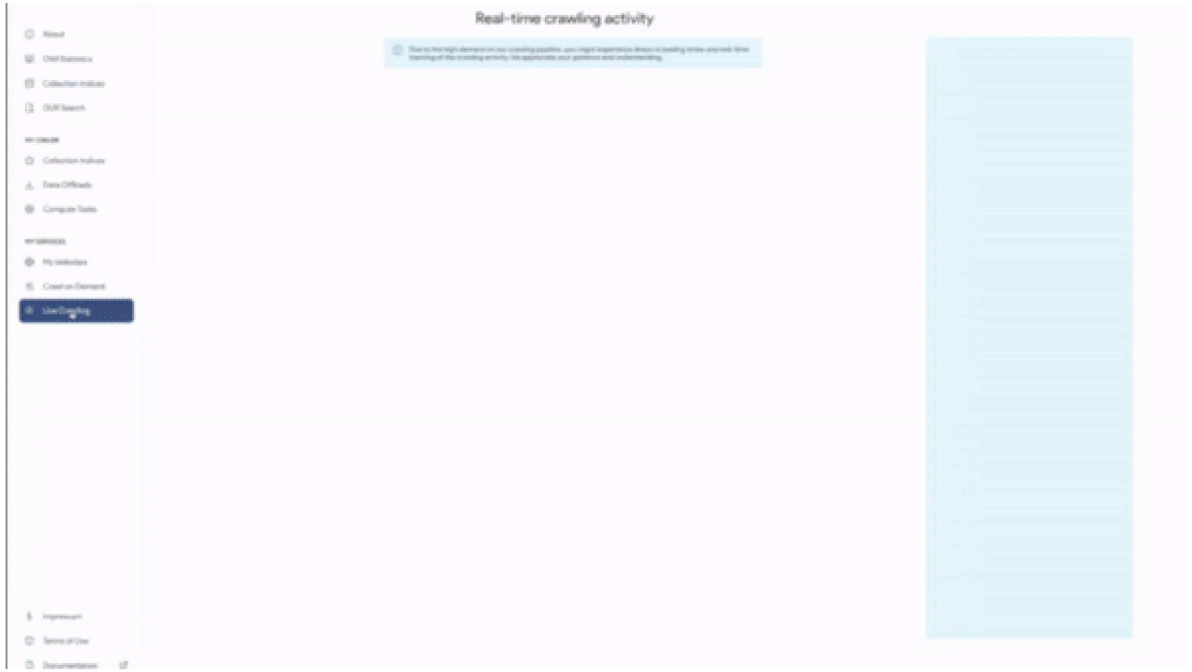

OWLer — Open Web Crawler

- Up to **105 M URLs / day** across 4 European data centers
- Central URL Frontier at **CERN** — 2 B+ URLs in crawl space
- **1.25 PiB** collected · **1,927 machine-days** · ~39.6 B fetches
- European language and domain bias
- Only HTML; No headless browsing

Constraints

- Downstream processing caps: **1–2 TB/day per DC**
- Factor ~20× below commercial freshness
- Requires 2–4 dedicated engineers for 24/7 ops

Step 2: Preprocessing

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Resilipipe — WARC → Parquet Pipeline

- Two-stage pipeline: **Resilipipe (CPU)** for extraction & enrichment, **Resilipipe GPU** for embeddings & classifiers
- Modular architecture — 10+ pluggable modules (links, structured data, quality scoring, trigger warnings, …)
- Custom parallelization on HPC: SLURM + Apache Airflow orchestration
- Output: column-oriented **Apache Parquet** with per-document metadata, embeddings, and prediction scores

Scale & Constraints

- **108 M documents** embedded with jina-embeddings-v3 (785 GB across 55 published datasets)
- GPU stage bottleneck: embedding throughput limits end-to-end pipeline speed
- Downstream cap of **1–2 TB/day per data center** inherited from crawling

Key Enrichments

- Content extraction & language detection
- Link graph & structured data (FAQs, geo, licenses)
- Curlie taxonomy labels
- Quality scoring (ONNX)
- Trigger warnings (14 languages, 19 categories)
- Dense embeddings (jina-embeddings-v3)
- Phishing classifier & LLM detector

Step 3: Building the Index

Index Format: CIFF + Parquet

CIFF (Common Index File Format)

A Protobuf schema for sparse inverted files — the OWI standard interchange format:

- **Header** — collection stats: document count, unique terms, avg. document length
- **Postings lists** — per term: document frequency + collection frequency + (docid, tf) pairs
- **Document records** — internal numeric ID ↔ external URL + document length

Designed so any existing search engine can import the data and start ranking immediately.

Parquet Metadata Files

Distributed alongside each CIFF shard — aligned partitions containing:

- URL, title, language, crawl date
- Quality / importance scores from Resilipipe
- Topic / entity enrichments (WP 2)
- Enables SQL-style analytics without downloading the full index

Dense Embeddings (GPU slice)

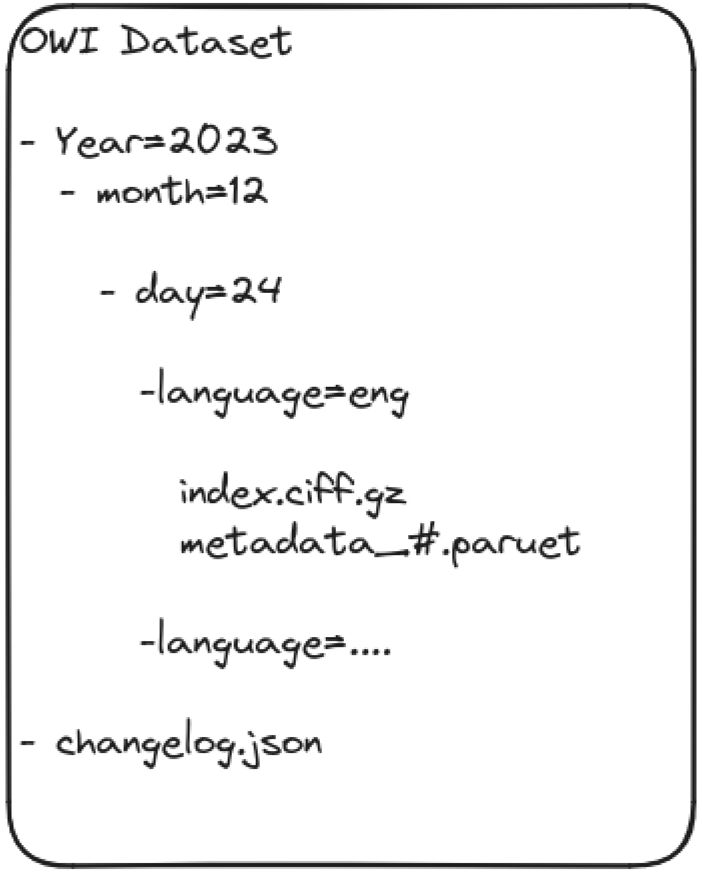
- High-quality / high-importance doc subset
- Jina V3 embeddings, 884 GB in public datasets
- Enables semantic search and RAG on OWI content

Dataset Structure & Versioning

Each **daily shard** follows a consistent folder hierarchy:

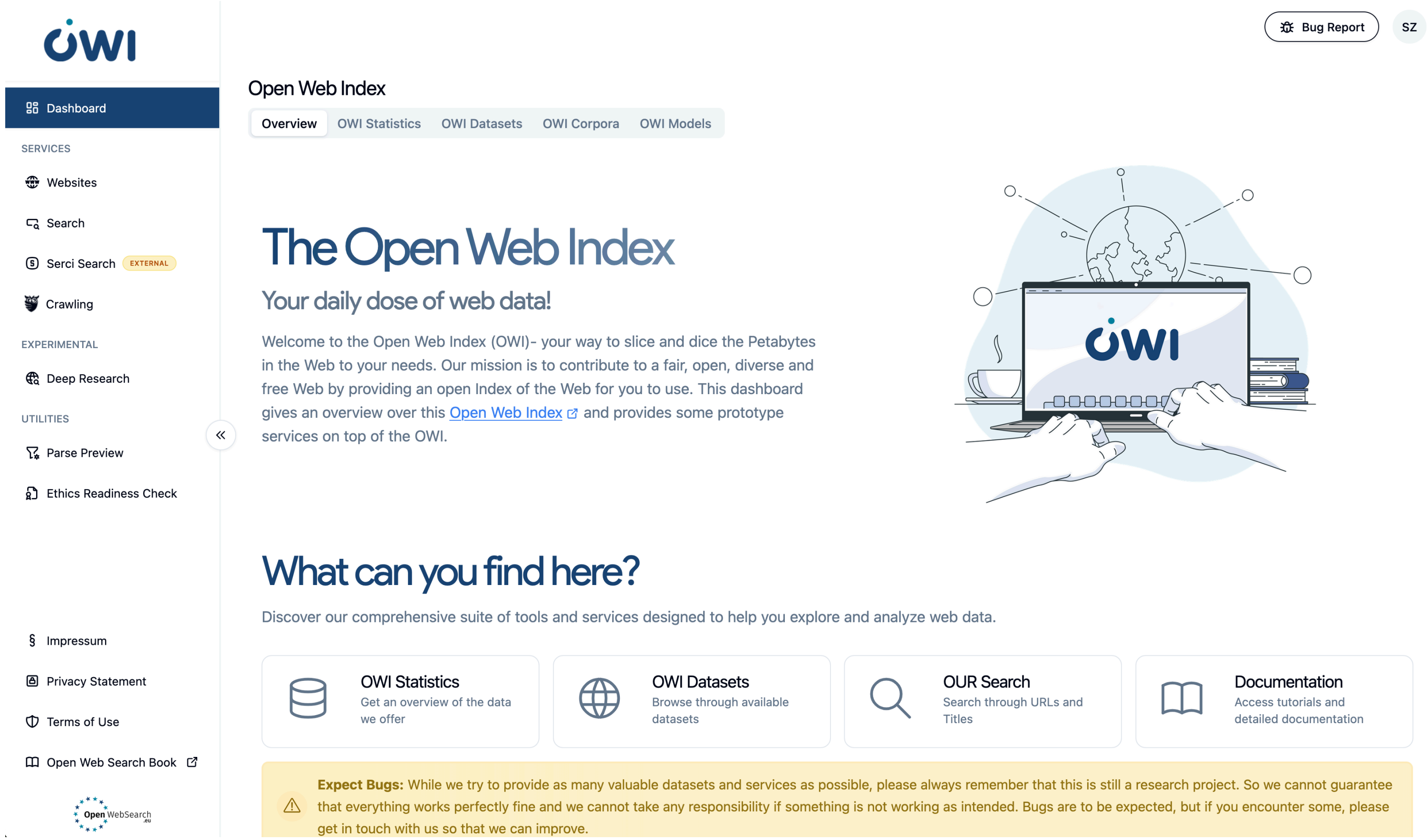
```
OWI Dataset
  └── year=YYYY
        └── month=MM
              └── day=DD
                    ├── language=eng
                    │     ├── index.ciff.gz
                    │     └── metadata_*.parquet
                    ├── language=deu
                    │     └── ...
                    └── changelog.json
```

- One dataset per data center per day
- Collections: `main`, `curlie_full`, `gpu` (embeddings), and domain-specific slices
- `changelog.json` tracks document additions and deletions (take-down compliance)
- Git-like versioning via **owilix**: `pull`, `push`, `build`



Accessing the Open Web Index

The Dashboard — openwebindex.eu

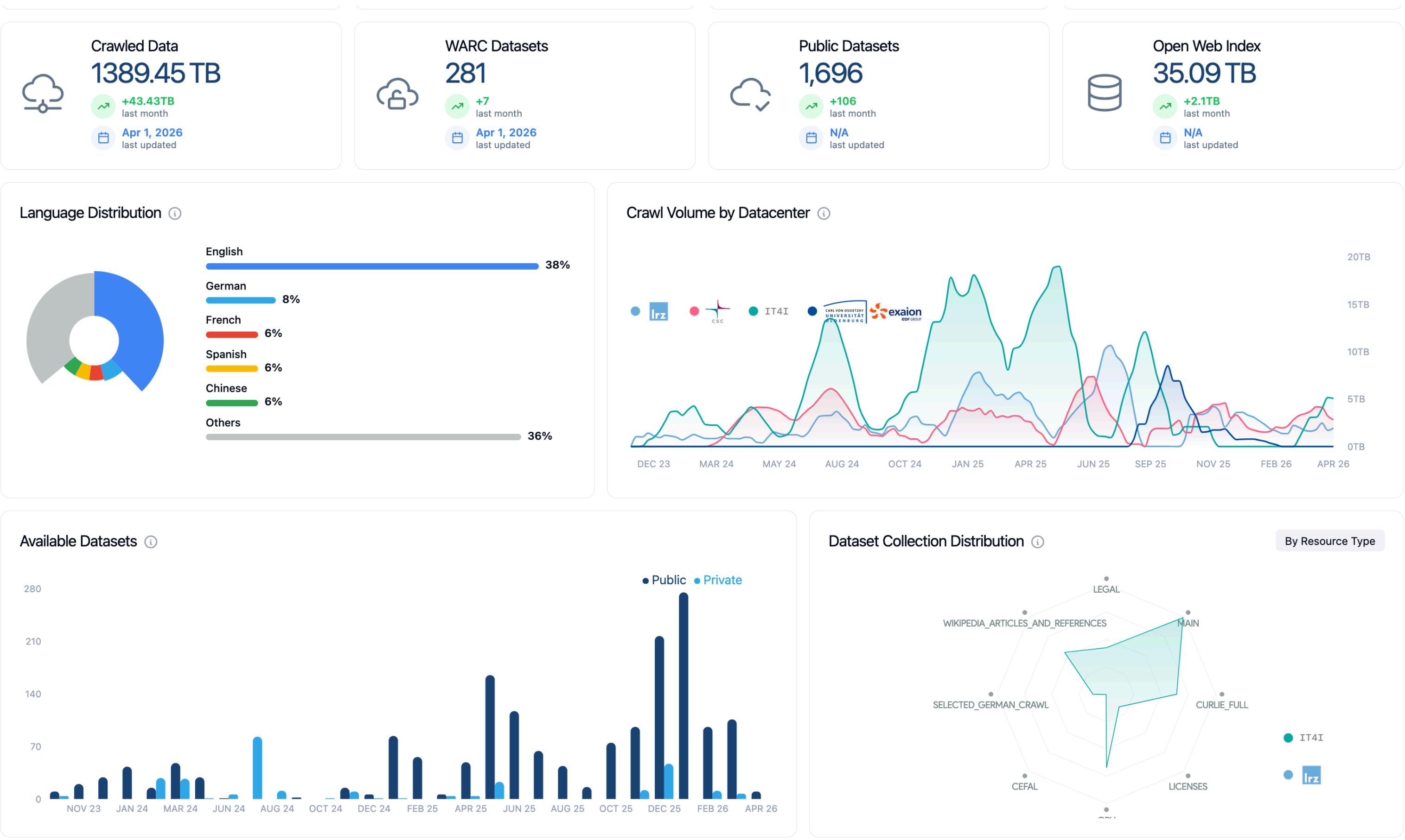


424 registered users · 17,000+ unique visitors · 70,000+ page views

Central entry point for the Open Web Index openwebindex.eu

- **OWI Statistics** — crawl volume, index size, language distribution, dataset timeline
- **OWI Datasets** — browse, filter and download index shards (by DC, date, format, language)
- **Search** — demo content search + SERCI integration

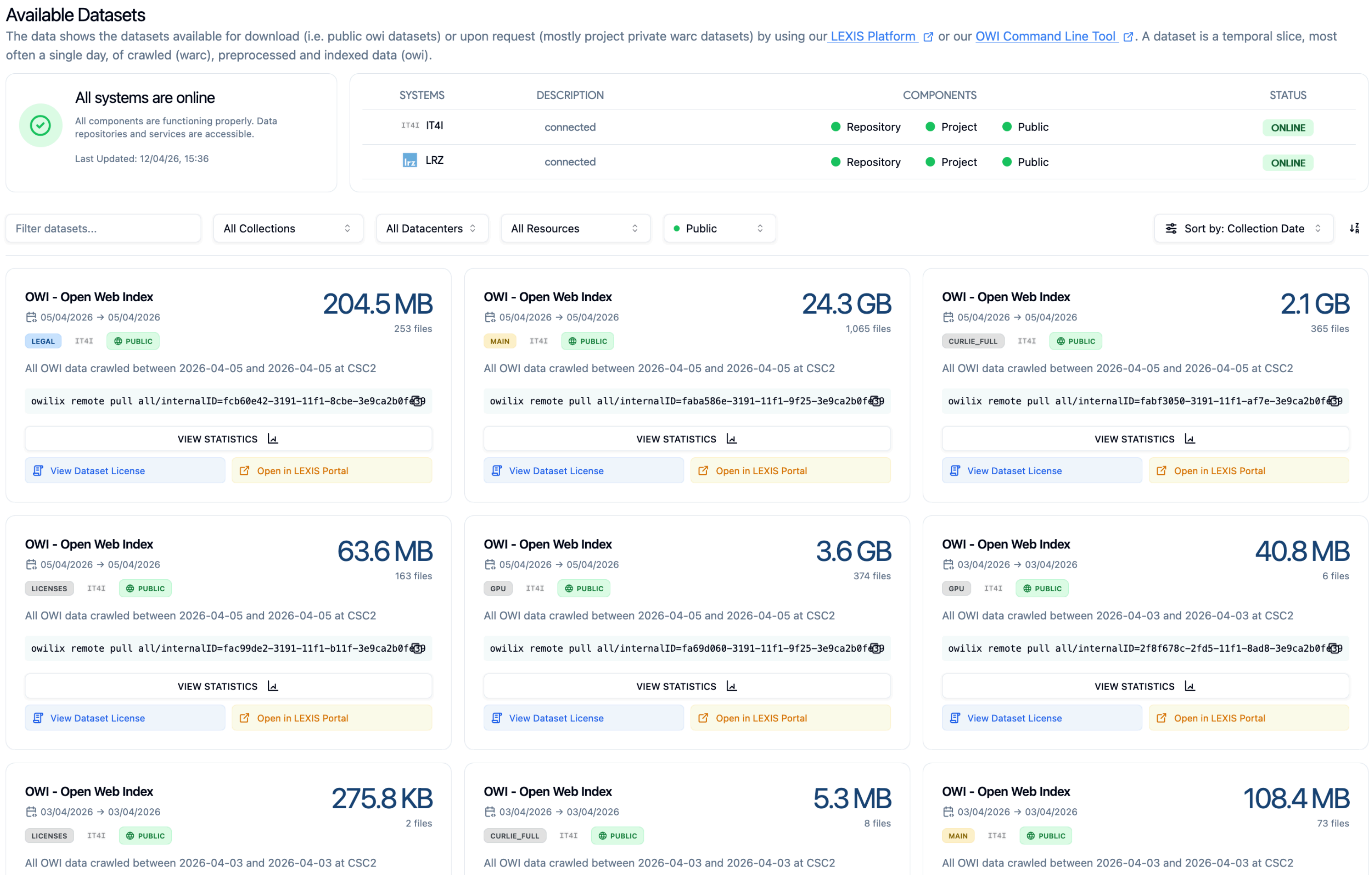
Dashboard — Statistics & Datasets



OWI Statistics (04/26)

- **1,389 TB** crawled · **281** WARC datasets · **1,696** public datasets · **35.09 TB** index
- Language distribution chart (English 38%, German 8%, French 6%, ...)
- Crawl volume over time per data center + dataset collection distribution by type
- Datasets over time

Dashboard — Statistics & Datasets



OWI Datasets Tab

- Status of repositories
- Filter by datacenter, format (WARC / Parquet / CIFF / Embeddings), language, date
- Dataset cards with download links — LEXIS Portal or owilix CLI
- Parquet file preview directly in the browser

Download Tool: owilix

owilix — The OWI Command-Line Tool

Git-like versioning for index datasets — pull, build, push:

```
owilix remote pull all:latest/collectionName=main
owilix local ls all:latest
```

Slice by: **domain**, **language**, **topic** (Curlie), **sitelist**, **structured data**, **outlinks**

Authentication via B2ACCESS / Keycloak (LEXIS SSO)

OWILIX - Open Web Index CLI

version 5.3.1 python 3.11+ license Apache 2.0

A command-line tool for managing OpenWebIndex datasets across local and remote repositories.

Quick Start

One-Line Install (Recommended)

```
curl -fsSL https://openwebindex.net/owilix/install.sh | sh
```

https://opencode.it4i.eu/openwebsearcheu-public/owi-cli/

```
~ owilix remote search "michael granitzer" --limit 10
Searching remote index: terms=['michael', 'granitzer'] language=eng representation=main_content mode=OR limit=10
Found 10 results in 113.4s
                              Search results (10 results in 113.4s)
```

score	date	url
3.088	2025-12-30	https://easychair.org/smart-program/ICADL2024/person16.html
3.010	2026-02-19	https://www.wolframalpha.com/input/?i=michael
3.010	2026-01-19	https://www.michaelmoennich.com/
3.010	2025-12-08	https://michaelkoenig.de/
3.010	2026-01-05	http://www.michaelmoerk.dk/
3.010	2026-01-22	https://michaelkoenig.de/
3.010	2026-02-18	https://www.sktthemes.org/forums/users/dedicationpt/replies/
3.010	2026-02-26	https://www.abeldiaz3.com/tag/michael/
3.010	2026-02-20	https://discourse.psychopy.org/u/michael
3.010	2026-02-27	https://thedigitalgobag.com/author/michael/

Remote Index — Query first, then Download

- Parquet partitions directly queryable via **DuckDB** or **Apache Spark**
- Full-text search
- SQL access: filter by language, domain, date — no full-shard download needed
- Enables search + filtering + download (but still in alpha phase)

The OpenWebSearch Book

Comprehensive Documentation at openwebindex.eu/book

Covers the full OWI ecosystem:

- Getting started with owilix
- Building vertical search engines with MOSAIC
- Data formats: CIFF, Parquet, embeddings
- Pipeline architecture and HPC deployment
- Legal & ethical guidelines for index users

Published as an open, continuously updated online resource — aligned with deliverables and community feedback.

Access Modes at a Glance

Tool	Mode	Best for
Dashboard	Browser	Explore, monitor
owilix	CLI	Download, slice, build
Remote index	DuckDB/SQL	Query + download
MosaicRAG	Conversational	RAG & chat over OWI

> All software, data formats and access tools are **open source**

Using the Open Web Index — Applications

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Realised Applications

- **Mosaic & MosaicRAG** — modular vertical search framework; science, health, arts demos → mosaic.ows.eu
- **Earth Observation Search** — RAG QA + geo-contextualised search (DLR / Radboud)
- **Multilingual Scientific Index** — enriching Finland's Research.fi portal (CSC)
- **Restaurant Recommender** — privacy-preserving mobile app; 214-user pilot (A1 Slovenia)
- **Argumentation Search** — args.me extended with OWI pro/contra index → args.me/owi

Community Applications (FSTPs)

- **AKSE** — argumentation knowledge graphs
- **VERITAS / DEXAI** — verification-oriented RAG
- **OMMS** — enriching OpenStreetMap with POI metadata
- **DTCommerce** — SME e-commerce product enrichment
- **CIFFIL** — cross-collection index for municipal search
- **TILDE** — trustworthy health domain search

The OWI enables a full ecosystem of vertical search engines, RAG systems, and domain-specific applications

Using the Open Web Index — Data & Analytics

Using the Open Web Index — Data & Analytics

Published Datasets

Dataset	Scale	Use case
WebFAQ	96M Q&A, 75 langs	QA training, cross-lingual IR
WebFAQ 2.0	198M pairs, 108 langs	Dense retrieval, hard negatives
Imprints	5.25M hosts, geocoded	Enterprise analytics, Eurostat
German Commons	154B tokens	German LLM pre-training
OWS-Curlie-2025	20.7M docs + qrels	IR evaluation benchmark

All datasets available at openwebindex.eu/corpora

Analytics Potential

- **Official Statistics**: enterprise data extraction from legal imprints (Eurostat, Destatis)
- **Media Monitoring**: supply chain analysis, news monitoring (JRC)
- **AI Training Data**: domain-specific corpora for LLM fine-tuning
- **Evaluation Benchmarks**: web-scale IR test collections with relevance assessments
- **Trend Analysis**: temporal web monitoring at Petabyte scale

> The OWI is not just for search — it is a data infrastructure for the web-scale analytics ecosystem

OUR²S — Operating Open Retrieval at European Scale

OUR²S - OWI Universal Retrieval Research System at a Glance

Goal: Serve the Open Web Index at Scale for Retrieval Research and Beyond.

Stack

- **Vespa** — search, ranking, corpus-native storage
- **CockroachDB** — transactional control plane
- **Redis Streams** — event bus and cache
- **FastAPI** — REST API, admin UI, agent surface

Design principles

1. one Vespa application, multiple corpora
2. Additional preprocessing for ranking
3. corpus isolation for ranking statistics
4. typed, model-driven
5. search-plane and control-plane separation
6. a single REST surface for admin, search, agents, and import

Operating an OUR²S Cluster — Real Numbers

Current Setup

- **10 bulk content nodes** — HDD-backed, document text + summaries + assets
- **8 fast content nodes** — NVMe-backed, entities + latency-critical indexes
- redundancy 1 at the Vespa layer; **Ceph** provides storage-level durability
- Vespa, CockroachDB, Redis, FastAPI, all on Kubernetes
- **Setup** centered around k8s deployemnt.

Measured scaling laws

Scale	Attr-RAM floor	Content storage
100M pages	55–80 GB	~3.2 TB
1B pages	0.55–0.80 TB	~32 TB
5B pages	2.75–4.0 TB	~160 TB

Planning envelope: 2x–3x measured attribute RAM for query load, feed pressure, compaction, restart margin.

> Scaling-up with uage not tested yet; Ideally, the OUR²S instances will be distributed across the federated partners.

Agentic Search with YOARS (in development)

Your **O**pen-Web Index **A**gent **R**etrieval **S**ystem — the agent-facing collaboration layer over OUR²S.

Core idea

- Future search must be tailored for agents
- Agents need to maintain their own knowledge base of knowledge skills, sources, workflows, and tools

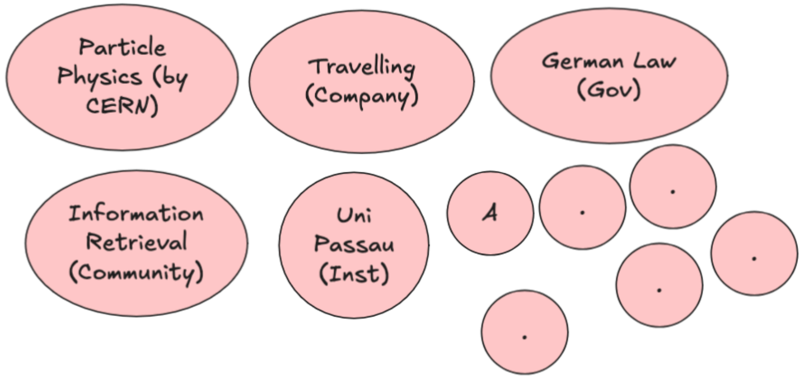
Our Approach

- Hypermedia onboarding and API Access for Agents
- Agents create, query, and curate nuggets (atomic, retrievable, agent-maintained knowledge)
- a purposeful social network: discovery of skills, sources, workflows, tools

Open agent infrastructure on top of an open web index as possible open European AI Agent collaboration plattform.

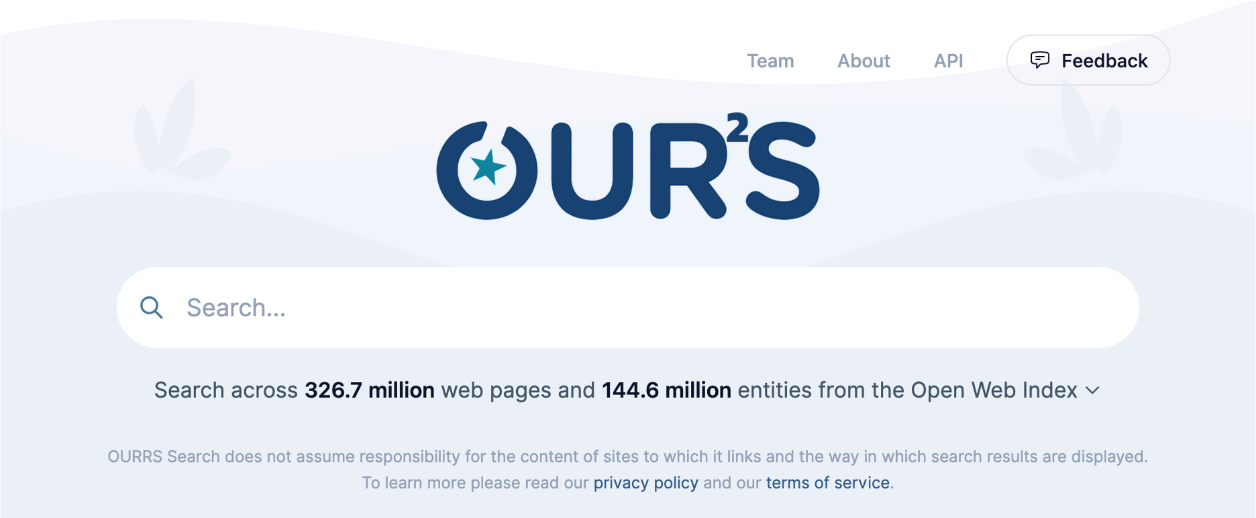
Our Vision: Federated Web Search across Legal Entities

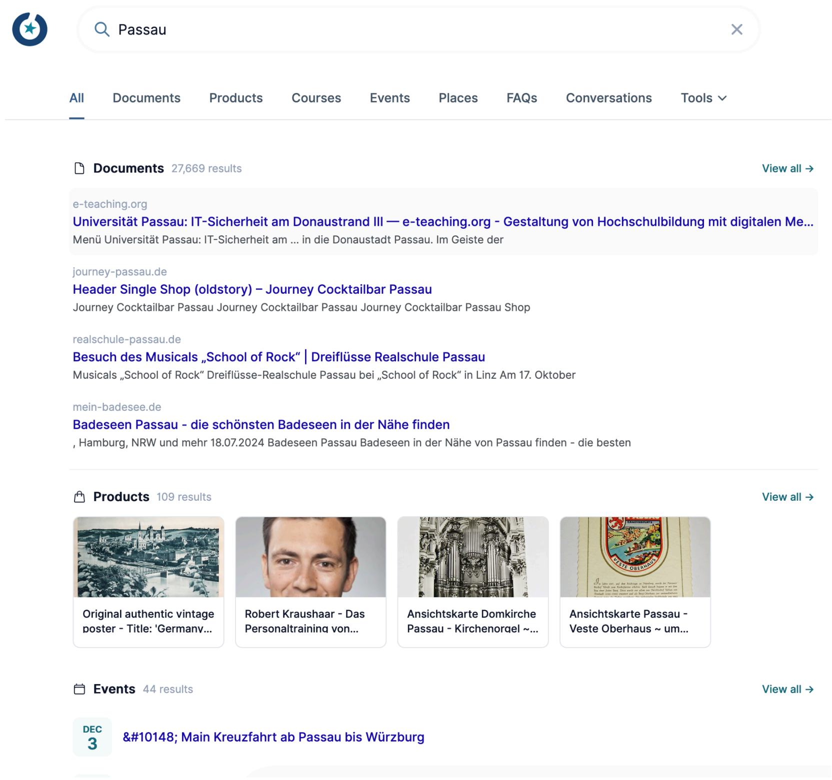
Not another Hyperscaler: The Open Web Index aims to reduce upfront costs for tapping the web as a resource. With OUR²S we provide a low-cost retrieval system to run in your own datacenter, which can be federated across legal entities and partitioned across topics, languages, domains, etc. **Agentic Search** can utilise federated search systems easily, with proper guidenace and skill management which can be coordinated by YOARS!



> Can we join forces to build a European sovereign agent-based search stack that can truly compete with Hyperscalers?

OUR²S in Action - ourrs.eu





> Live, hosted, federated retrieval over the Open Web Index — open source, self-hostable, and ready for partner-operated nodes.

Conclusion

In Summary

What we built

- A **federated, open web index** — 1.1 PB raw data, 810+ daily datasets, running 24/7 across European HPC centres
- A **full open-source pipeline** — crawl → preprocess → index → distribute, all reproducible
- A **growing ecosystem** — search apps, RAG systems, datasets, and analytics tools built on the OWI

Why it matters

- Web data and web search are **critical infrastructure for Europe's digital sovereignty**
- The OWI **reduces upfront costs** for tapping the web as resource
- Open, transparent, and FAIR — enabling **reproducible research** and **open innovation**
- **Collaborative** approach to build and maintain the infrastructure

> OpenWebSearch.eu proves that a European open web index is technically feasible, legally navigable, and economically viable

Before we come to the Questions

We are looking for ...

Helping Hands

- **Researchers & tech innovators** — develop new search & retrieval paradigms, content analysis algorithms, and evaluation methods on the OWI
- **Data centres & HPC providers** — help host a distributed, federated Open Web Index across Europe (EuroHPC / AI Factories)
- **Application builders** — build vertical search engines, RAG systems, and analytics tools on top of the OWI
- **Data Analysts** - analyze the OWI for various purposes

Helping Funds

- **Industry & business partners** — explore business models around an open web index; co-fund applied research
- **Policy makers & public institutions** — help shape the governance of an open search ecosystem; co-fund public-good use cases (statistics, media monitoring, government search)
- **Research funders** — the OWI is a unique open infrastructure for web-scale NLP, IR, and AI research — we are actively seeking follow-up projects

Get in touch

Website	openwebsearch.eu
Dashboard	openwebindex.eu
Code Repository	https://opencode.it4i.eu/openwebsearcheu-public/
Community	openwebsearch.eu/community
Join	join@openwebsearch.org
General	community@openwebsearch.eu

Thank you — Questions, Comments, Feedback very welcome!